Md Nazmul Islam

1840 rhodes road, Kent, Ohio 44240

+12342829758 | nazmul87@gmail.com | nazmulislam-lab | nazmul87 | https://nazmulislam-lab.github.io/

Research Interests

Robotics, Imaging system, Quantum Sensor, Computer vision, Human-Machine Interface, Control system, Machine learning, Deep learning, Healthcare AI, Explainable AI, , Biomedical signal analysis, Data mining, IOT.

Education

Kent state University *Kent, Ohio, USA*

PHD IN MECHATRONICS, ROBOTICS AND AUTOMATION ENGINEERING *August, 2024 - continuing*

- CGPA: 4.00/4.00
- Academic Standing: NA
- Thesis: NA
- Advisor: Dr. Tao Shen

BRAC University *Dhaka, Bangladesh*

MASTER OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING *April, 2019 - February, 2022*

- CGPA: 4.00/4.00
- Academic Standing: Highest Distinction
- Thesis: Demystify the Blackbox Model of Automated Detection of Lung and Kidney Diseases from X-ray and CT Radiographs
- Advisor: Dr. Md Golam Rabiul Alam

Chittagong University of Engineering and Technology (CUET) *Chittagong, Bangladesh*

BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING *April, 2007 - May, 2011*

- CGPA: 3.18/4.00
- Merit Position: 46th
- Thesis: Energy and power Pricing Analysis of Bangladesh 2010-2011
- Advisor: Dr. Shamsul Alam

Publications

JOURNAL PAPER

Md Nazmul Islam, Md. Golam Rabiul Alam, Tasnim Sakib Apon, Md. Zia Uddin, Nasser Allheeib, Alaa Menshawi, Mohammad Mehedi Hassan. 2023. Interpretable Differential Diagnosis of Non-COVID Viral Pneumonia, Lung Opacity and COVID-19 Using Tuned Transfer Learning and Explainable AI. MDPI Healthcare, 11(3), 410, 31 January 2023.

Md Nazmul Islam, Mehedi Hasan, Md. Kabir Hossain, Md. Golam Rabiul Alam, Md Zia Uddin, Ahmet Soylu. 2022. Vision Transformer and Explainable Transfer Learning Models for Auto Detection of Kidney Cyst, Stone and Tumor from CT-Radiography. Nature Scientific Reports, 12(1), 11440, 06 July 2022.

CONFERENCE PAPER

Kylie Schwartz, **Md Nazmul Islam**, Jarif Ahmed, Tao Shen. 2026. A Universal Joint-Adaptive Orthosis with Velocity Modulated Resistance. ASME Frontiers in Biomedical Devices, V001T04A005, 30 April 2026.

Izel Tuncer, **Md Nazmul Islam**, Renhao Jin, Tao Shen. 2025. Universal Joint Orthosis (UJO): Assistive and Resistive Movement Device for Astronauts and Medical Rehabilitation. ASME Frontiers in Biomedical Devices, V001T07A003, 30 April 2025.

Anand, **Md Nazmul Islam**, Renhao Jin, Qiang Guan, Yan Sun, Gengqing Song, Tao Shen. 2025. Development of a Wearable Device for Assisting Diaphragmatic Breathing. ASME Frontiers in Biomedical Devices, V001T04A005, 30 April 2025.

Md Nazmul Islam, Mehedi Hasan, Abdul Kadar Muhammad Masum, Md Zia Uddin, Md. Golam Rabiul Alam. 2021. Demystify the Black-box of Deep Learning Models for COVID-19 Detection from Chest CT Radiographs. 2021 24th International Conference on Computer and Information Technology (ICCIT), IEEE, pp. 1–6, 18 December 2021.

Personal Details

Date of Birth : 01-Sep-1989

Nationality : Bangladeshi

Current Location : Dhaka, Bangladesh

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, MATLAB

AI / ML: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Vision Transformers, Transfer Learning, Explainable AI (XAI)

AI Agents & LLM Tools: Claude, Codex, Gemini, Grok Bot, Cursor, OpenClaw, Hermes Agents, multi-agent orchestration, local LLMs

Web & Apps: React, Vite, Socket.IO, Pixi.js, Node.js, GitHub Pages

Robotics & Hardware: Arduino, Proteus, wearable and assistive device prototyping, Hugo robotic surgical system

Medical Imaging Systems: CT, PET-CT, O-arm and navigation systems (Philips, Medtronic)

Professional Experience

August, 2024 -present	**Graduate Assistant, Kent state University**, **Duties/Responsibilities :** 1) Research quantum sensors and medical robotics. 2) Teach the Mechatronics Lab and the Signal and Electronics Lab.
2021-August, 2024	**Field Technical Service Specialist, Hugo Surgical Robotics, Medtronic**, **Duties/Responsibilities :** 1) To give field service support for Hugo Robotic systems. 2) Responsible for Installation and commissioning. 3) Responsible for uptime of Hugo Robotic system. 4) Coordination of logistics related to Hugo spares. 5) Up-keeping of Demo Hugo system. 6) Regularly visit existing customers and maintain network for post sales service etc. to reinforce the relationship and understand service gaps. 7) Plan for spares inventory to cater to the services of Hugo Robotics in ISC region. 8) Responsible for completion of any FCA and quality guidelines from BU 9) Maintaining the dashboard of service activities. 10) Responsible for O-arm and Navigation machine
2013-2021	**Field Service Engineer, CT and PET CT - Philips HealthCare**, **Duties/Responsibilities :** 1) Installation, Preventive Maintenance, Modification upgrades, Trouble shooting and Repair of CT/AMI machine. 2) Installation, Preventive Maintenance, Modification upgrades, Trouble shooting and Repair of Multi modality Workstation(ISP,EBW). 3) Maintain service records/ reports and maintain compliance. 4) Assist to Achieve CP%, Revenue and AR as per regional target.
2012-2013	**Executive Engineer, MRI and X-ray, Siemens Bangladesh Ltd.** , **Duties/Responsibilities :** Servicing, Maintenance and Installation of MRI and support for heliophos-D, and orthophos (Sirona)
2012-2013	**Lecturer, EEE (Part-Time), Daffodil International University**, **Duties/Responsibilities :** Taking lab and classes of Digital Signal Processing, Electrical Circuit, Electronic circuit-1

Certifications

Academic Courses Certification- Workera

Machine Learning Researcher, Issued by: Workera, March 2022.

Academic Courses Certification-Coursera

Convolutional Neural Networks | DeepLearning.AI

Convolutional Neural Networks in TensorFlow | DeepLearning.AI

Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization | DeepLearning.AI

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning | DeepLearning.AI

Structuring Machine Learning Projects | DeepLearning.AI

AI for Medical Diagnosis | DeepLearning.AI

Probability and Statistics: To p or not to p? | University of London

Python Data Structures | University of Michigan

Professional Certification

Hugo RAS(Robotic Assisted Surgery) training , North Haven, Connecticut, USA, 2nd November 2021.

Remote Technical Field Service Training for the Hugo RAS System,June 2021

LEAN Basic Training at Philips Healthcare, Mumbai, India, August 2018.

PET-CT(Ingenuity TF PET CT differences) at Philips Healthcare Academy, Cleveland, the USA , September 2017

PET CT(NM Gemini TF Series PET Table GSU with Truflight Select),Philips Healthcare Academy, Cleveland, the USA , Sep. 2017.

Intellispace Portal R10 Server at Gurgaon Philips, India, August 2018.

CT Scanner (Brilliance Air Family) at Singapore Learning Centre, Philips Singapore, July 2015.

CT Scanner(Brilliance/Ingenuity/iCT Gateway) at Singapore Learning Centre, Philips Singapore , July 2015.

CT Scanner(CT advance training MX-16) at Professor Holst Centre, the Netherlands, August 2014.

O-arm imaging technical training , Hyderabad, India, February 2022.

References

Reference-01 Dr. Tao Shen
Associate Professor
School of Engineering, College of Aeronautics and Engineering
Kent State University
Kent, Ohio, USA
e-mail: tshen3@kent.edu

Reference-02 Dr. Md Golam Rabiul Alam
Professor
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
e-mail: rabiul.alam@bracu.ac.bd

Reference-03 Sanjay Guha Ray
Proactive Monitoring Engineer - CT
Philips Healthcare
Kolkata, India
e-mail: sanjay.guharay@philips.com